

Akshitha Theretupally

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PROFESSIONAL SUMMARY

Master's student in Computer and Information Sciences (Dec 2025) with 2+ years of experience in full-stack software engineering across enterprise, e-commerce, and academic settings. Skilled in Java (Spring Boot), Python, React, SQL, and cloud platforms (AWS, Azure) with strong foundations in data structures, algorithms, and object-oriented programming. Experienced in building scalable web applications, APIs, and cloud-native systems while collaborating cross-functionally with product and design teams. Passionate about Nextdoor's mission to strengthen local communities and eager to apply AI and emerging technologies to reimagine user experiences. Authorized to work in the U.S. for 3 years without sponsorship (F-1 OPT + STEM OPT).

EDUCATION

Master's in Computer and Information Sciences, George Mason University, GPA: 3.92
B.Tech in Computer Science, CMR Institute of Technology, GPA: 3.68

Dec 2025
May 2022

EXPERIENCE

Toyz Electronics

May 2025 - Aug 2025

E-Commerce Full Stack Developer Intern

Remote

- Shipped new APIs and SQL-backed features used in production within first weeks, improving data consistency by 20%.
- Developed JUnit/Mockito tests for microservices, raising automated test coverage to 80% and improving reliability.
- Contributed to CI/CD with Jenkins and OpenShift, reducing build verification time by 30%.
- Refactored backend services for AWS Lambda + RDS, improving scalability for high-traffic e-commerce operations.
- Collaborated with design and product teams to scope requirements and ensure smooth user experiences.

George Mason University

Mar 2024 - July 2025

Learning Space Design Assistant

Fairfax, VA, USA

- Built ASP.NET MVC + SQL Server apps to manage 50+ classrooms, cutting reporting mismatches by 50%.
- Modeled OOP-based backend logic and optimized SQL queries, enhancing system responsiveness.
- Automated validation scripts, reducing manual verification workload by 40%.
- Authored SOPs, test cases, and documentation to streamline onboarding and ensure project continuity.
- Supported migration of reporting systems to Azure Functions + Blob Storage, modernizing infrastructure.

DBS Tech

Jul 2022 - Dec 2023

Application Developer

Hyderabad, TG, India

- Developed scalable Spring Boot + Angular applications in an Agile team, improving system performance.
- Designed and deployed REST APIs, optimized ETL pipelines, and improved downstream analytics.
- Monitored services with Elasticsearch + Kibana, enabling faster incident resolution.
- Partnered with QA/DevOps to automate CI/CD pipelines on Jenkins/OpenShift, accelerating deployments.
- Worked closely with product managers and stakeholders to align features with business goals.

PROJECTS

- **Graph Algorithm Visualizer (Java, OOP, DSA):** Implemented Kruskal's, Prim's, and Bellman-Ford algorithms with object-oriented design, enabling step-by-step visualization of graph traversal and shortest path computations.
- **Database-Backed User Management System:** Developed a user management system with CRUD functionality (create, read, update, delete) backed by a relational database. Implemented authentication logic, role-based access control, and efficient query handling using SQL indexes. Designed schema to support scalability and optimized queries for fast lookups.
- **AI-Powered Sentiment Analyzer (Python, NLP):** Built a sentiment analysis tool using scikit-learn + NLTK to classify 10k+ customer reviews into positive, negative, and neutral categories. Improved model accuracy by 12% via feature engineering (TF-IDF, n-grams).
- **Neighborhood Chatbot (Python, OpenAI API, React):** Built an AI-powered chatbot prototype that provides neighborhood-level updates (e.g., events, alerts, FAQs) by integrating GPT-based responses with structured local datasets. Developed a React frontend and connected it to backend APIs for real-time responses and interactive user experience.

SKILLS

- **Programming & CS Fundamentals:** Java, Python, C#, Data Structures, Algorithms, OOP
- **Web & App Development:** Spring Boot, ASP.NET MVC, Node.js, React, Angular, REST APIs
- **Databases:** SQL Server, PostgreSQL, MySQL, Oracle, MongoDB
- **Cloud & DevOps:** AWS (Lambda, RDS, S3, EC2, CloudWatch), Azure (Functions, Blob Storage), Jenkins, Docker, Kubernetes, OpenShift
- **Testing & QA:** JUnit, Mockito, Selenium, Jasmine, TestNG, TDD/BDD practices
- **Collaboration & Tools:** Git/GitHub/GitLab, Jira, Confluence, Agile/Scrum

CERTIFICATIONS

- **AWS Certified Cloud Practitioner:** Amazon Web Services, 2025
- **MTA: Introduction to Programming Using Python:** Microsoft, 2020